

ENTERPRISE KUBERNETES MASTERY

AI Platform Engineering Handbook

PART XIV ENTERPRISE REFERENCE ARCHITECTURES

Financial Services, Healthcare, Government, AI Platforms, RAG, Agentic AI

Volume 14 of 16 Architecture Series | Edition 2025-2026

CHAPTER 1

Enterprise Kubernetes Platform -- Full Stack Architecture

The following reference architecture represents a production-grade enterprise Kubernetes platform meeting security, compliance, observability, and operational requirements for regulated industries. Each layer is independently selectable and all components are CNCF-graduated or widely adopted in production.

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LAYER 8: AI PLATFORM GPU Operator (NVIDIA) + MIG Manager + DCGM Exporter Volcano (gang scheduling) + Karpenter (GPU node provisioning) vLLM (LLM serving) + KServe (multi-framework) + Ray Serve LiteLLM (AI gateway) + MLflow (registry) + Kubeflow Pipelines Feast (feature store) + Qdrant/Milvus (vector DB) Temporal (durable agent execution) + Argo Workflows (batch AI) LAYER 7: DEVELOPER PLATFORM Backstage (IDP: service catalogue, templates, docs) ArgoCD + ApplicationSets (GitOps, multi-cluster) Crossplane (self-service infrastructure) Argo Rollouts (progressive delivery) LAYER 6: OBSERVABILITY Prometheus + Thanos (metrics, long-term S3) Fluent Bit -> Loki (logs, S3 backend) OTel Collector -> Tempo (distributed traces) OpenCost + DCGM (cost + GPU observability) Grafana (unified dashboards) LAYER 5: SECURITY OIDC/Dex (human auth) + SPIFFE/SPIRE (workload identity) Vault HA + External Secrets Operator (secrets) cert-manager + Vault PKI (certificate lifecycle) Kyverno (admission policy) + OPA Gatekeeper (complex Rego) Falco (runtime detection) + Tetragon (enforcement) Cosign + Trivy + Harbor (supply chain security) LAYER 4: NETWORKING Cilium eBPF CNI (kube-proxy replacement + WireGuard) Istio Ambient Mesh (mTLS, L7 policy, zero sidecar) ingress-nginx (external) + Gateway API (internal routing) CoreDNS + NodeLocal DNSCache LAYER 3: STORAGE AWS EBS gp3 / GCP PD-SSD (block, RW0) EFS/NFS / Ceph CephFS (shared, RWX) Velero (backup, DR) LAYER 2: KUBERNETES CONTROL PLANE API Server (private endpoint, audit log -> S3 immutable) etcd (3 nodes HA, encryption at rest, 30-min backup) Volcano scheduler (AI) + default scheduler (standard) LAYER 1: INFRASTRUCTURE Managed cloud (EKS/GKE/AKS) or self-managed (CAPI) CPU node pools + GPU node pools (A100/H100) Private VPC, no public node IPs Cloud IAM + HSM for key management
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Financial Services Reference Architecture

Requirement	Control	Implementation
Data sovereignty	No PII leaves jurisdiction	Cluster per region; Kyverno blocks external registries; no cross-region data flow
Encryption at rest	All data encrypted with customer keys	etcd AES-GCM + KMS; PVC LUKS; S3 SSE-KMS; Vault CMEK
Encryption in transit	All comms encrypted	Cilium WireGuard (L3) + Istio Ambient mTLS (L7) + cert-manager
Privileged access	MFA + break-glass audit	OIDC + Okta MFA; Vault lease-based admin; every action audited
Vulnerability SLA	CRITICAL: 24h; HIGH: 7d	Trivy CI gate + Harbor daily scan + Kyverno image age policy

Requirement	Control	Implementation
Change management	All changes via approved GitOps	ArgoCD selfHeal + prune; no kubectl write in prod; PR required
Business continuity	RTO < 4h; RPO < 1h	Velero hourly; multi-AZ cluster; etcd 30-min snapshot to S3
AI model audit (EU AI Act)	Article 12: complete audit trail	OTel GenAI traces -> Tempo; MLflow lineage; immutable S3 audit log
PCI DSS v4	Cardholder data isolation	Dedicated namespace + cluster; NetworkPolicy + Vault for card data

Healthcare Reference Architecture

Healthcare Kubernetes deployments must satisfy HIPAA, HITRUST, and increasingly FDA 21 CFR Part 11 for AI/ML medical devices. Key additions to the base platform:

- * **PHI isolation:** Dedicated namespace per patient data category. Kubernetes NetworkPolicy: no cross-namespace communication for PHI workloads. Encryption at rest with FIPS 140-2 validated modules.
- * **De-identification pipeline:** AI-powered PII/PHI detection (Presidio) deployed as admission webhook: blocks any attempt to log raw PHI to non-compliant systems. All PHI hashed/tokenised before AI model processing.
- * **Audit trail:** Every access to PHI data logged with: user identity, timestamp, purpose (clinical/research/admin), data elements accessed. Immutable audit log in compliance-certified object storage.
- * **Medical AI governance:** FDA 21 CFR Part 11: electronic records and signatures for AI model validation, versioning, and deployment approval. MLflow + custom approval workflow CRD for pre-market submission artifacts.

Enterprise Agentic AI Platform Reference Architecture

Full-stack enterprise agentic AI platform (derived from Part XII):

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REQUEST ENTRY API Gateway (Kong) -> WebSocket / REST endpoint Authentication: OIDC JWT validation Rate limiting: per user, per team, per cost center AI GATEWAY LAYER LiteLLM (unified model API) Model routing: complexity-based (8B for simple, 70B for complex) Cost tracking: token usage per session per user PII detection: Presidio sidecar before external LLM calls AGENT ORCHESTRATION Temporal Worker Pool (KEDA-scaled on Kafka lag) Agent Registry (Agent CRD + Backstage catalogue) Prompt Registry (ConfigMap + GitOps versioning) Tool Registry (AgentTool CRD + MCP server routing) AGENT EXECUTION Stateless Workers (Deployment, 2-100 replicas, KEDA) Code Execution (Kata Containers, gVisor, isolated namespace) MCP Servers (shared Deployment + KEDA on call rate) MEMORY LAYER Redis (working memory, TTL-based session context) Qdrant (semantic/vector memory, RAG retrieval) PostgreSQL + TimescaleDB (episodic memory, audit log) Feast (entity profiles, user features) LLM INFERENCE LAYER vLLM (production LLM serving, PagedAttention) NVIDIA GPU Operator + MIG (GPU resource management) KServe (multi-framework, canary model deployments) OBSERVABILITY OTel GenAI traces: every LLM call, tool call, agent step TTFT, TPS, queue depth, cost/session in Prometheus Agent decision log in Loki (compliance retention 1 year) Grafana AI platform dashboard
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